

So the best thing is to change the entire production line, and this can be done very easily. Many companies have already done that.

Apple is the biggest example. Nokia and Sony have also done that. They have changed their entire production line. Any of Apple products, be it MacBook or iPod, doesn't contain brominated compound or PVC. Their products are on the verge of becoming fully green and have started reaping profits for them.

Success stories

Apple has done some basic design improvement in its MacBook by removing the high-current cable. A laptop has two kinds of cable wires providing electricity. One is the thicker cable which is directly connected to the socket point and then there is the adaptor and a thinner cable. The thicker cable is made of PVC. Apple removed it so that the adaptor can directly plug in. It saved them cost and in the process they also turned green. This saves cost at the end of the product's life too, as the payment for recycling goes down.

Recycling cost depends on what kind of toxic material you use in your product and in what amount. Toxic chemicals like mercury, PVC and cadmium can't be recycled. These have to be treated and disposed of. If you don't use them in your products, it saves you money in production and your payment for recycling too goes down.

Nokia has started using cellulose plastics in its products, which is biodegradable. So there is no cost incurred in treating these. Sony Ericsson has removed all the brominated compounds from its products.

Binay Opto Electronics, a manufacturer of LED illumination products in India, has come up with PowerLED dome-type high-bay luminaries that eliminate annual bulb and ballast replacement costs as well as hazardous disposal costs. These luminaries are eco-friendly as there is no mercury and lead used, and are a long-life re-



Recycling electronics helps reduce pollution (Image courtesy: <http://techcrunch.com>)

placement for existing lamps in indoor high-bay lighting in factory sheds, industrial shop floors, warehouses, etc. They offer low power consumption and a maintenance-free life in an economical form factor.

On account of the 180°-directivity property of LEDs, which enables the LEDs to be directed where required in the workspace, no reflector is required in the luminaire itself. This ensures that there is no loss of luminous energy in reflection. The luminaire saves 65-70 per cent energy over comparable metal-halide or HPS lamps, and lasts 10 to 15 years.

Way to go

"I don't see there is much difficulty for companies in greening the products. It's only about the intent and commitment. If companies can sell a phone in places where there is not even a road, they can do it too. The government is very supportive.

If you look at the current e-waste legislation which is under implementation, it is very progressive in nature and the best-in-class at global scale. And as far as the value chain (suppliers and all) is concerned, if you give them a bulk order for green components, they will manufacture it," says Pratap.

For those willing to go green, Pratap suggests: "First, you have to comply with the legislation currently in place. Second, you have to see green not as a compulsion but as a business opportunity. You have to see green as sustaining your business in the long term. When all around the environment is changing with the resources depleting, you have to go for innovation and find out how you can sustain the material supply in a continuous manner. Solution lies in the use of the recycled material rather than going for virgin material." ●

The author is executive editor for EFY magazine